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**UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER 2, 2024/2025**

SUBJECT CODE : SECI1143 / SCST1223

SUBJECT NAME : PROBABILITY & STATISTICAL DATA ANALYSIS

SECTION : 01 - 08

TIME :

DATE/DAY :

VENUES :

INSTRUCTIONS:

- Answer **ALL** questions in the given **ANSWER BOOKLET**.
- This paper consists of **SIX (6) QUESTIONS**:
- DO ALL CALCULATION IN 3 DECIMAL POINTS.**
- Submit **BOTH Questions and Answer Booklet** at the end of the exam.

(Please Write Your Lecturer Name and Section in Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This question paper consists of **NINE (9)** printed pages including this page.

STRUCTURED QUESTIONS (100 MARKS)

INSTRUCTION:

This section consists of **SIX (6)** questions. Answer **ALL** questions in the answer booklet.

QUESTION 1

[10 MARKS]

- a) A bioinformatic researcher is studying the effect of a new medication on cholesterol levels. A sample of 10 patients was collected and their cholesterol levels before taking the medication were recorded. After 6 months, the following reduction in cholesterol levels (in mg/dL) were observed as in **Table 1**.

Table 1: Reduction in Cholesterol Levels (mg/dL) After 6 Months of Medication

17, 12, 19, 14, 13, 15, 16, 18, 13, 14
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It is known from past studies that the population standard deviation is 2.5 mg/dL. Answer the following questions based on the given case study:

- i) Using the 10 patients' data, calculate the sample mean as a point estimate for the population mean reduction in cholesterol levels. (1 mark)
- ii) Construct a 97% confidence interval for the population mean reduction in cholesterol levels. Based on your interval, determine whether there is sufficient evidence to support the claim that the average reduction exceeds 10 mg/dL. (4 marks)
- b) A social media influencer claims that their new 2-week digital detox challenge reduces average daily screen time among university students to below 6 hours per day. To test this claim, a university student group participates in the detox challenge. After 14 days, a sample of 12 students report an average daily screen time of 5.4 hours, with a sample standard deviation of 1.2 hours. Assume that the daily screen time is normally distributed. At the 0.05 significance level, test the hypothesis whether the average daily screen time after the detox challenge is indeed less than 6 hours. (5 marks)

QUESTION 2**[20 MARKS]**

- a) A health and wellness company are comparing the effectiveness of two types of workout programs (Program X and Program Y) on customer weight loss results. They collected weight loss data (in kilograms) from two independent samples of customers who followed either Program X or Program Y for three months. The goal is to determine if there is a significant difference in the average weight loss between the two programs. A summary of the data is shown in **Table 2**.

Table 2: Data on Weight Loss Results for Two Different Workout Programs

	Number of participants, n_i	Mean weight loss, $\bar{x}_i$ (kg)	Standard deviation, s_i (kg)
Program X	45	6.8	1.5
Program Y	55	7.4	1.2

- i) State the hypothesis statement. (2 Marks)
- ii) Calculate the test statistic. (2 Marks)
- iii) Calculate the degree of freedom. (2 Marks)
- iv) At $\alpha = 0.05$ level of significance, and assuming equal variances, test whether there is a significant difference in average weight loss between the two programs. (4 Marks)

- b) A nutritionist wants to test whether a new diet plan improves cholesterol levels in individuals. A group of patients has their cholesterol levels measured before and after a 6-week diet plan. The results are shown in **Table 3**.

Table 3: Cholesterol Levels Before and After a 6-Week Diet Plan

Patient	A	B	C	D	E
Before diet	220	210	225	215	230
After diet	200	198	205	190	210

- i) State the null and alternative hypotheses. (1 Mark)
- ii) By using a significance level of $\alpha = 0.05$, does the diet plan significantly reduce cholesterol levels? (7 Marks)
- iii) Based on the result of the hypothesis test in part (ii), interpret the outcome in terms of the effectiveness of the diet plan. (2 Marks)

QUESTION 3**[20 MARKS]**

- a) A game developer wants to test if a new virtual wheel spinner in their mobile game is fair. The wheel has 5 equal segments, each representing a reward: Gold, Gems, Energy, Tokens, and Keys. They spin the wheel 100 times and record how often each reward appears. The results are presented in **Table 4**.

Table 4: Frequency of Rewards from 100 Spins of the Virtual Game Wheel

Reward	Frequency
Gold	14
Gems	19
Energy	25
Tokens	22
Keys	20

The developer wants to know: Is the wheel spinner fair, or are some rewards appearing more often than expected by chance?

- i) What are the expected frequencies for each reward segment? (1 Mark)
- ii) Calculate the difference between observed (o) and expected frequencies (e) for each reward segment. (5 Marks)
- iii) Conduct the test at the 0.05 significance level to evaluate the developer claim. (6 Marks)

- b) A media company is curious to understand whether gender influences people's choice of streaming platforms. To explore this, they conduct a survey at a large university campus during a *Tech and Entertainment Fair*.

The survey asks participants to select their most-used streaming platform among four popular services: Nx, D+, PV, and HMx. Participants also indicate their gender. The goal is to determine whether platform preference is dependent to gender. The summarized responses from 120 students (split evenly between male and female) are clearly illustrated in **Table 5**.

Table 5: Summary of Responses from 120 Students by Gender

	Nx	D+	PV	HMx	Total
Male	30	10	10	10	60
Female	20	20	10	10	60
Total	50	30	20	20	120

At the 0.05 significance level, determine whether there is a statistically significant association between gender and preferred streaming platform? (8 Marks)

QUESTION 4**[15 MARKS]**

A researcher wants to investigate whether daily social media usage (measured on a scale of 1-10, where 10 = most frequent) affects sleep quality (measured on a scale of 1-10, where 10 = best sleep). Data from ten participants is collected as depicted in **Table 6**.

Table 6: Daily Social Media Usage and Sleep Quality of Ten Participants.

Participants	Social Media (X)	Sleep Quality (Y)
1	2	8
2	4	6
3	1	9
4	3	7
5	5	5
6	2	8
7	6	4
8	1	9
9	3	7
10	4	6

- a) Calculate the Spearman's Rho Rank Correlation (r_s) for the data collected in **Table 6**.
Provide your answer in 3 decimal places. (13 Marks)
- b) Based on the Spearman's Rho Rank Correlation (r_s) result from (a), does there appear to be a linear relationship between daily social media usage and sleep quality? How would you characterize the relationship? (2 Marks)

QUESTION 5**[15 MARKS]**

The ages, t years of 7 patients and their blood pressure, p mmHg are shown in **Table 7**.

Table 7: Blood Pressure of 7 Patients

Patient	A	B	C	D	E	F	G
t	42	74	48	35	56	26	60
p	98	130	120	88	182	80	135

Based on the given data,

- i) Sketch a scatter diagram of the data. (3 marks)

- ii) Give the reason to support fitting a regression model to these data. (2 marks)

- iii) Using the least squares equation, calculate the intercept of the regression line, rounding your answer to three decimal places (5 marks)

- iv) Find the linear regression equation of p on t . (2 marks)

- v) Estimate the blood pressure of a 40-year-old patient. (3 marks)

QUESTION 6**[20 MARKS]**

Students prepared for an exam using three different study methods: group study, online tutorials, and self-study. Their exam scores are summarized in **Table 8**. Conduct a hypothesis test to determine whether there are significant differences in the mean scores among the different study methods.

Table 8: Comparison of Exam Scores Across Different Study Methods

Group Study (A)	Online Tutorials (B)	Self-Study (C)
78	85	72
81	88	70
75	82	68
80	90	74

Based on the given data,

- i) Define the hypothesis null, H_0 and hypothesis alternative H_1 . (2 marks)
- ii) Calculate the mean and standard deviation for each category. (6 marks)
- iii) Calculate the variance between samples. (3 marks)
- iv) Calculate the variance within samples. (1 mark)
- v) Find the value for test statistic. (2 marks)
- vi) Calculate the numerator and denominator degree of freedom. (2 marks)
- vii) Find the critical value, with $\alpha = 0.05$. (2 marks)
- viii) Test the claim and state the conclusion. (2 marks)

-END OF QUESTIONS PAPER-